Human-in-the-Loop Network Intrusion Detection with a Hybrid CNN-LSTM, Uncertainty-Aware Routing and Controlled Retraining

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***Abstract*—Benchmark scores of learning-based intrusion detectors say little about what happens once live traffic drifts away from what the detector was trained on. This paper describes a working prototype built around that gap. Each network flow is first labelled by a convolutional-recurrent (CNN-LSTM) classifier. A routing stage then decides whether the verdict can be acted on directly or must be checked by a security analyst; it combines a temperature-scaled entropy score, a Mahalanobis distance measured on the classifier's 128-dimensional embedding, and the vote spread and agreement of a Random Forest acting as a second opinion. Analyst verdicts, with their stated reasons and evidence, are kept in a searchable incident store, and they influence the model only after an administrator has verified them and a retrained candidate has cleared thresholds declared before training began. We evaluate the prototype on the corrected release of CIC-IDS2017, after removing identifier columns and collapsing 385,106 duplicated flows before the data were split. On 257,222 untouched test flows the deployed configuration forwards 1.48% of traffic for review; that queue holds 98.9% of the classifier's mistakes, and the traffic handled without a human is labelled with 99.999% accuracy, with two attack flows dismissed and no benign flow escalated. Withholding one attack family at a time shows that the classifier's own confidence collapses on novel attacks (up to 97% silently dismissed), whereas the combined router keeps such losses at or below 10.2% for five of the six withheld families. Brute-force SSH logins escaped every configuration we tested.**

***Index Terms*—intrusion detection, human-in-the-loop, uncertainty estimation, CNN-LSTM, out-of-distribution detection, CIC-IDS2017, selective classification**

I. Introduction

Our starting point was a simple observation from our own first experiments: a Random Forest trained in a few minutes already scored 99.97% on a random split of CIC-IDS2017. If a quick baseline can reach that figure, accuracy on such a split cannot be what separates a usable detector from an unusable one. Three things weaken numbers of this kind. Flow datasets carry mislabelled and repeated records that leak between training and test partitions [3], [4]. Columns that identify hosts or sessions let a model memorise rather than learn [5]. Above all, a deployed detector will sooner or later receive an attack family absent from its training data, and a softmax output gives no reliable signal when that happens [6], [11].

Our design follows an uncertainty-aware adaptive IDS proposed in [1], which pairs a CNN-LSTM with cWGAN-GP augmentation and analyst feedback and reports 98% binary accuracy, 96.6% detection rate, 92.1% precision and 94.3% F1. We rebuilt its classifier and turned the idea into a complete application. Rather than asking how accurate the classifier is, we ask a more operational question: for which flows may the classifier act alone, and which ones must a person look at before anything is closed?

The paper makes three contributions.

1) We compare the CNN-LSTM with Random Forest and LightGBM on the corrected CIC-IDS2017 data under strict leakage control (shortcut columns removed, duplicates collapsed before splitting, three seeds), together with ablations that remove the LSTM or add synthetic samples.
2) Using withheld attack families and a split by capture day, we show that the network's own uncertainty overlooks most novel attacks, and we propose a combined routing rule (entropy, embedding distance, and a tree-ensemble second opinion) that sends most of them to an analyst.
3) We implement the full loop as software: traffic replay, an analyst console that captures the reason behind each verdict, an incident store with similarity search, and retraining that is gated by thresholds fixed in advance, administrator approval, live deployment and rollback.

II. Related Work

Our system draws on four strands of earlier work: the benchmark data and its known defects, classifiers for tabular flow records, ways of estimating when a classifier should not be trusted, and generative oversampling. Rather than restate each source, Table I lists what we took from it and where our system departs from it.

III. Dataset and Preprocessing

All experiments use the corrected CIC-IDS2017 files [3], [4], one file per capture day from Monday to Friday. Together they hold 2,099,976 flow records, 91 columns and 27 distinct label strings. Our inspection script found no empty cells and

TABLE I
PRIOR WORK USED IN THIS SYSTEM

Source	Idea relevant to us	How this project uses or extends it
[2]	Five-day labelled flow capture (CIC-IDS2017)	Our only training and test source
[3], [4]	Flow and label errors found; corrected release published	We use the corrected files and add our own de-duplication before splitting
[5]	Leakage, shortcuts and weak baselines distort security ML	Identifier columns and destination port dropped; tree baselines on identical inputs
[6]	Real networks are open-world; unseen activity is the norm	Motivates withholding whole attack families during evaluation
[1]	CNN-LSTM, cWGAN-GP and analyst feedback for adaptive IDS	Classifier rebuilt; feedback loop made controlled, verified and reversible
[7]	Gated recurrent memory	Recurrent layer over convolutional feature maps
[8], [9]	Bagged and boosted tree ensembles	Baselines; the Random Forest also serves as second opinion
[10]	One temperature parameter rescales logits	Fitted on half of the validation data
[11]	Softmax confidence flags many errors	Entropy score used as one routing signal
[12]	Distance to class centres in feature space flags unfamiliar inputs	Applied to the LSTM embedding
[13]	Independent models disagree on unfamiliar inputs	Disagreement between CNN-LSTM and Random Forest triggers review
[14]	Trade abstention rate against error on accepted inputs	Referral rate vs. error on auto-handled flows
[15], [16]	Class-conditional GAN with gradient penalty	Minority oversampling ablation

only five records with infinite byte- or packet-rate values. No two rows were identical as stored, yet once the identifier columns were set aside, one row in eight (12.5%) turned out to be a repeat.

Columns removed. Flow ID, both IP addresses, the source port, the timestamp and the row counter describe *who* talked, not *how*, so the model never sees them. The corrected release adds an "Attempted Category" column that is only filled for attack rows; it would hand the answer to the model and is discarded as well. We also leave out the destination port: in this capture, port 21 is almost synonymous with FTP-Patator, which a model could exploit without learning anything about the attack itself.

Label groups. Every "Attempted" variant (an attack that did not succeed) is folded into its parent attack. The *Infiltration–Portscan* label is folded into *Portscan*, since the two are indistinguishable in feature space, and brute-force, XSS and SQL-injection web attacks form one "Web Attack" class. The result is 13 classes. Heartbleed (11 records) and Infiltration (81 records) are kept but flagged: their per-class scores rest on a handful of test flows, so we also report a "reliable" macro-F1 computed without them.

Removing repeats before the split. Rows that agree on every input feature and on the label are reduced to one copy, which removes 385,106 rows; a further 58 rows whose identical features carry two different labels are dropped because no classifier could separate them. Excluding the destination port exposed how repetitive the scan traffic is: Portscan shrank from 230,833 to 7,695 distinct flows. The 1,714,812 remaining flows are divided 70/15/15 with class stratification (Table II), and hashing every feature vector confirms that no flow occurs in more than one partition.

Feature transform. Every statistic below is estimated on the training partition alone. Infinite and missing values are replaced by the training median. Because byte and duration counts span many orders of magnitude, each value is passed through $\operatorname{sign}(x)\log(1+|x|)$. One column that never varies and 17 columns that almost duplicate another ($|\rho| > 0.99$) are dropped, leaving 64 inputs, which are then standardised. The fitted transform travels with every saved model together with the column order and class list, so the web application applies exactly the processing used in training.

TABLE II
FLOWS PER CLASS AFTER MAPPING AND DE-DUPLICATION

Class	Train	Validation	Test
BENIGN	996,410	213,517	213,516
DoS Hulk	111,333	23,857	23,858
DDoS	66,601	14,272	14,271
Portscan	5,386	1,154	1,155
DoS GoldenEye	5,353	1,147	1,147
DoS Slowloris	3,992	855	856
DoS Slowhttptest	3,576	766	766
FTP-Patator	2,789	597	598
SSH-Patator	2,092	448	448
Web Attack	1,439	309	308
Botnet	1,343	288	288
Infiltration†	46	10	10
Heartbleed†	8	2	1
Total	1,200,368	257,222	257,222

†Too few samples for reliable per-class metrics.

IV. METHODOLOGY

A. Classifier

The layer sequence is taken from Table 5 of [1], with 64 inputs instead of 46. The standardised feature vector is read as a single-channel signal of length 64 and passes through a 64-filter convolution (kernel 5, ReLU), max-pooling by two, a 128-filter convolution (kernel 3, ReLU), an LSTM with 128 units, dropout of 0.3 and a 13-way linear output. We keep the last LSTM state $\mathbf{z} \in \mathbb{R}^{128}$ as an embedding of the flow, since later stages measure distances in that space. Training uses ordinary cross-entropy with AdamW (learning

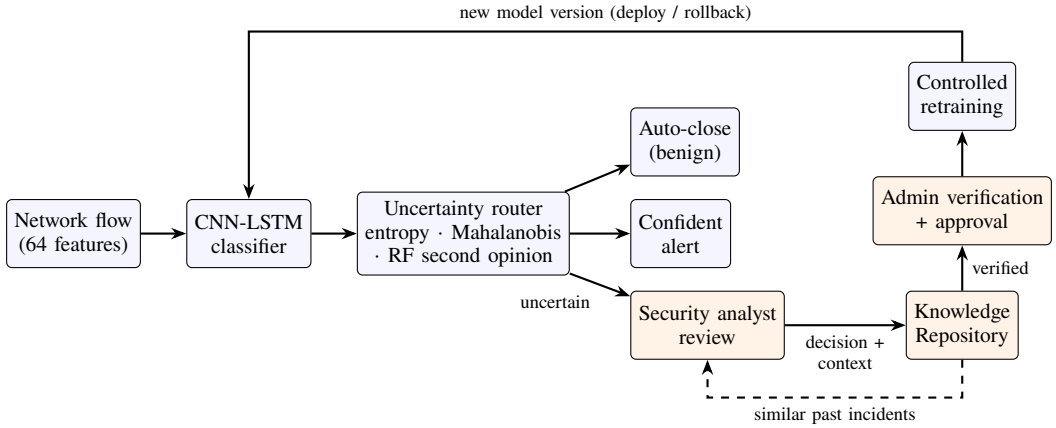


Fig. 1. System overview. Only uncertain flows reach the analyst; verified analyst decisions improve the model only through administrator-approved retraining.

rate 10^{-3}, batches of 1,024, cosine decay), up to 20 epochs, and keeps the epoch with the best validation macro-F1. We avoided class-weighted loss because it shifts the predicted probabilities that the routing stage relies on. Our first runs, however, never once predicted Heartbleed or Infiltration, which held validation macro-F1 near 0.85; copying the rows of any class with fewer than 1,000 training examples until it reaches that count (training partition only) fixed this.

B. Baselines and ablations

A 100-tree Random Forest and a LightGBM model receive exactly the same 64 transformed inputs. LightGBM stops early on validation loss; in our first attempt it diverged after 12 boosting rounds, which a cap on leaf output values resolved. Three ablations probe the design: replacing the LSTM with global average pooling, adding GAN-generated minority samples, and skipping temperature scaling.

C. Minority augmentation

For the augmentation ablation we train a class-conditional Wasserstein GAN with gradient penalty [15], [16]. Generator and critic each have two hidden layers of 256 units; the penalty weight is 10 and the critic is updated five times per generator update. Only classes with between 100 and 5,000 training rows are topped up, to 5,000 rows each. A generated row is accepted only when the baseline Random Forest assigns it to the class it was generated for. The reference design seeds its generator with a VAE; ours starts from Gaussian noise.

D. Calibration and uncertainty signals

We cut the validation partition into two stratified halves. The first ("calibration") half is used to fit everything in this subsection; the second ("evaluation") half is reserved for judging the result. A single temperature T [10] rescales the logits, after which every flow receives two scores:

$$H(\mathbf{x}) = -\frac{1}{\log K}\sum_{k=1}^{K} p_k \log p_k, \quad \mathbf{p} = \mathrm{softmax}(\mathbf{f}(\mathbf{x})/T), \quad (1)$$

an entropy over the $K = 13$ classes scaled to lie between 0 and 1, and the squared Mahalanobis distance from the flow's embedding to the nearest class centre [12]

$$D(\mathbf{x}) = \min_c (\mathbf{z} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{z} - \boldsymbol{\mu}_c), \quad (2)$$

where the centres $\boldsymbol{\mu}_c$ and one pooled covariance $\boldsymbol{\Sigma}$ (with a small ridge added for stability) come from training-set embeddings. Entropy captures hesitation between known classes; the distance captures flows that resemble none of them. Each score gets a cut-off read from the calibration half: τ_H is the value exceeded by its most uncertain 1% of flows, and τ_D the value exceeded by its most distant 0.5%.

E. Hybrid routing

A flow is *uncertain* if

$$H > \tau_H \ \vee \ D > \tau_D \ \vee \ u_{\mathrm{RF}} > \tau_{\mathrm{RF}} \ \vee \ \hat{y}_{\mathrm{RF}} \neq \hat{y}, \quad (3)$$

and each flow receives one of three decisions:

$$\mathrm{decision}(\mathbf{x}) = \begin{cases} \text{review} & \text{if uncertain,} \\ \text{auto-alert} & \text{else if } \hat{y} \neq \text{BENIGN,} \\ \text{auto-benign} & \text{otherwise,} \end{cases} \quad (4)$$

Here $\hat{y}$ denotes the CNN-LSTM's predicted class and $\hat{y}_{\mathrm{RF}}$ the Random Forest's. The forest's own doubt is $u_{\mathrm{RF}} = 1 - \max_k p_k^{\mathrm{RF}}$, i.e. the share of trees that did not vote for the winning class, and τ_{RF} flags its most doubtful 0.5% of calibration flows. In words: a flow is closed or escalated automatically only when the network is decisive, the flow lies close to familiar traffic, and an independently trained forest is both confident and in agreement. We introduced the two forest terms only after the withheld-family experiment (Section VI-D) exposed how often the network alone dismissed novel attacks. That experiment therefore cannot count as independent evidence for the combined rule; the test partition, opened once and only after the rule was frozen, can.

F. Human-in-the-loop platform

Fig. 1 shows how the parts connect. The back end is a FastAPI service backed by SQLite; the front end is a React application with three areas: a public About page, an analyst console and an administration console.

- *Replay.* To imitate live traffic, a background worker feeds the untouched test partition to the active model in configurable batches. Each record's true label is stored beside it purely so the administrator can measure live accuracy; it never enters the model and is never returned to analyst screens, a property checked by an automated test.
- *Analyst console.* A referred flow opens on one screen showing its addresses, ports and time, the three most likely classes, the forest's opinion, both uncertainty scores drawn against their cut-offs, the six features furthest from typical training values, and the most similar earlier incidents. The analyst answers with a verdict (malicious, benign or unsure), a class, and free-text reason, context and evidence, plus a confidence level and tags.
- *Incident store.* Every answer is saved as a searchable incident. Similarity between incidents is the cosine of their CNN-LSTM embeddings; because a retrained network maps flows to a different space, the stored embeddings are recomputed whenever the active model changes.
- *Gated retraining.* Feedback becomes training data only after an administrator marks it as verified. The verified items are split 70/30. The active network is fine-tuned on the 70% together with 200,000 rows drawn from the original training partition, so that it does not forget older attack families, and is then recalibrated. Old and new models are compared on two fixed yardsticks: the evaluation half of the validation partition and the 30% of feedback held back. The pass mark is written in a configuration file and copied into each job when the job is created: reliable macro-F1 may fall by no more than 0.5 points, the false-positive rate may rise by no more than 0.05 points, and accuracy on the held-back feedback may not fall. Even a passing candidate goes live only after an administrator approves it; deploying or rolling back replaces the model in memory while the service keeps running, and each step is recorded in an audit log.

V. Experimental Setup

Models are fitted on the training partition, and every choice of settings or design was made by looking at validation results only. We opened the test partition a single time, once the routing rule and all settings were frozen. Each model was trained with three seeds (42, 7 and 2025), and tables give the mean and standard deviation across them; the deployed application uses the seed-42 network and forest. "Accuracy" counts a prediction as correct only if the exact one of 13 classes is right. Precision, detection rate (recall), F1, false-positive rate (FPR) and false-negative rate instead pool all attacks into one positive class, as the reference work does. Everything ran on a laptop with an Intel Core i5-1240P, 7.7 GB of RAM and an NVIDIA RTX 2050 (4 GB), using PyTorch [17] and scikit-learn [18].

Generalisation tests. Two further experiments use only training and validation rows. (i) *Withheld family:* for each of six attacks we delete every row of that attack from the data used for training and calibration, refit the whole chain with one seed (eight CNN-LSTM epochs, temperature, cut-offs and forest), and then record what the router does with that attack's flows. (ii) *Split by day:* we fit on Monday to Wednesday traffic and evaluate on Thursday and Friday, whose attack types never occur earlier in the week.

VI. Results

A. Classification on the held-out test set

Table III sets every model beside the figures published for the reference design. All five of our models, the rebuilt CNN-LSTM included, score above those figures on every binary metric. We do not read this as an improvement on that work: the data release, the duplicate handling and the removed columns all differ, and such choices alone are known to move results a great deal [3], [5]. Within our own comparison the tree ensembles come out ahead. The Random Forest reaches 99.23% reliable macro-F1 against the CNN-LSTM's 98.09%, and raises roughly a third as many false alarms (FPR 0.020% vs. 0.059%).

Looking class by class (Table IV), the network's shortfall is concentrated in three classes: Portscan, Botnet and Web Attack. Botnet is the weakest of the reliable classes for all three models, with recall between 89% and 95%.

B. Ablations

Taking the LSTM out costs 1.7 points of reliable macro-F1 and drops Portscan recall from 96.7% to 88.8%, so the recurrent layer does earn its place. Adding generated minority samples lifts mean reliable macro-F1 by 0.4 points and makes the three seeds agree more closely (standard deviation falls from 0.99 to 0.25); the gain is, however, smaller than the spread between seeds. A simple check explains why the effect is modest: a classifier asked to tell real from generated rows separates them perfectly (AUC 1.0) for all six topped-up classes, because real flow features are integer counts whereas the generator emits smooth real values. Temperature scaling settled on $T = 1.01$, meaning the network's confidence was already in line with its accuracy (calibration error 0.023% before and 0.022% after); dropping this step changes nothing measurable here.

C. Routing on the test set

Table V summarises the configuration that runs in the application. One flow in roughly 68 (1.48%) is put in front of an analyst, and that small queue contains 98.9% of all the mistakes the network makes. On the remaining 98.5% of traffic, which nobody reviews, the labels are correct 99.999% of the time. Out of 43,706 attack flows, two were closed as benign, and not a single benign flow was escalated as a

TABLE III
HELD-OUT TEST SET (257,222 FLOWS), MEAN ± STANDARD DEVIATION OVER THREE SEEDS (%)

Metric	Reported [1]	Random Forest	LightGBM	CNN-LSTM	CNN-LSTM + aug.	CNN (no LSTM)
Accuracy	98.0*	**99.97** ± 0.00	99.96 ± 0.00	99.89 ± 0.07	99.92 ± 0.02	99.80 ± 0.02
Detection rate	96.6	**99.92** ± 0.00	99.91 ± 0.01	99.75 ± 0.24	99.86 ± 0.05	99.23 ± 0.10
Precision	92.1	**99.90** ± 0.00	99.84 ± 0.01	99.71 ± 0.07	99.74 ± 0.03	99.70 ± 0.01
F1	94.3	**99.91** ± 0.00	99.88 ± 0.01	99.73 ± 0.16	99.80 ± 0.04	99.46 ± 0.05
Macro-F1 (13 classes)	–	**99.07** ± 0.22	98.32 ± 0.04	96.71 ± 3.06	97.46 ± 1.36	94.12 ± 1.42
Macro-F1 (reliable)	–	**99.23** ± 0.02	99.03 ± 0.05	98.09 ± 0.99	98.47 ± 0.25	96.39 ± 0.32
FPR	–	**0.020** ± 0.000	0.033 ± 0.003	0.059 ± 0.015	0.054 ± 0.005	0.062 ± 0.002

*Binary accuracy in [1]; ours is 13-class accuracy. "Reliable" excludes Heartbleed and Infiltration.

TABLE IV
PER-CLASS RECALL ON THE TEST SET (%, MEAN OVER THREE SEEDS)

Class	RF	LightGBM	CNN-LSTM
BENIGN	99.98	99.97	99.94
DoS Hulk	100.00	99.99	99.90
DDoS	100.00	100.00	99.99
Portscan	99.65	99.65	96.65
DoS GoldenEye	100.00	99.91	98.23
DoS Slowloris	100.00	100.00	99.10
DoS Slowhttptest	98.87	99.09	98.96
FTP-Patator	100.00	100.00	99.78
SSH-Patator	99.70	99.70	99.48
Web Attack	97.94	98.38	96.54
Botnet	94.79	92.01	89.35
Infiltration†	93.33	80.00	93.33
Heartbleed†	100.00	100.00	100.00

†10 and 1 test flows respectively.

TABLE V
DEPLOYED ROUTING ON THE TEST SET

	Hybrid (deployed)	CNN-LSTM signals only
Referred to analyst	1.48%	1.33%
Accuracy, auto-handled	**99.999%**	99.997%
Errors caught by review	**98.9%**	96.0%
Attacks auto-closed	**2**	7
Benign auto-alerted	0	0

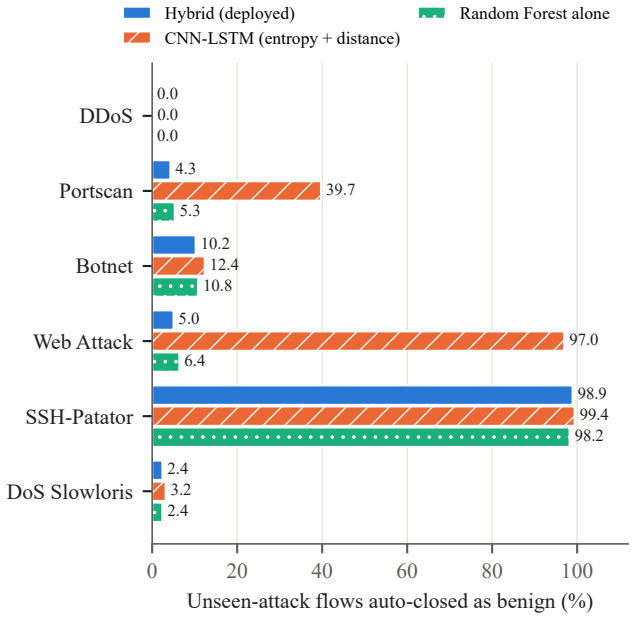


Fig. 2. Leave-one-attack-out: share of the held-out attack's flows auto-closed as benign (lower is better). Each group retrains the full pipeline without that attack.

confident alert. Against routing with the network's own two scores, the forest adds 0.15 percentage points of review work and cuts dismissed attacks from seven to two. The review load falls where the classifier is least reliable: every Infiltration flow, 99.7% of Web Attack, 75% of DoS Slowhttptest, 63% of Botnet and 38% of Portscan flows reach an analyst, compared with under 1% for BENIGN, DDoS and DoS Hulk.

D. Unseen attacks: leave-one-attack-out

For each withheld family, Fig. 2 plots the fraction of its flows that the router closed as benign, i.e. the attacks nobody would ever have seen. Our first pass of this experiment is what changed the design: relying on its own two scores, the network dismissed 80.5% of the unseen Portscan flows and 32.9% of the unseen Botnet flows, while a Random Forest that only looked at how its trees split their votes dismissed 5.3% and 10.8%. The network also behaved inconsistently from one training run to the next: unseen web attacks were dismissed 13.1% of the time in the first pass and 97.0% in the pass plotted here. With the combined rule, dismissals were lowest for five families out of six and never above 10.2% for those five; the price is that about 1.4% of ordinary traffic is reviewed, where the forest alone needs 0.55%. SSH brute force got past every variant: taken one at a time, its login attempts look exactly like genuine SSH sessions.

E. Unseen days

When the models see only Monday to Wednesday and are tested on Thursday and Friday, the combined rule closes 1.38% of attack flows as benign. Using the network's entropy alone, that figure is 35.28%; adding the embedding distance brings it down to 16.46%, so the distance score does real work, but only the forest's opinion brings losses down to the level of the forest by itself (1.45%). Ordinary traffic sent for review rises to 0.76%, against 0.50% for the forest alone. Broken down by

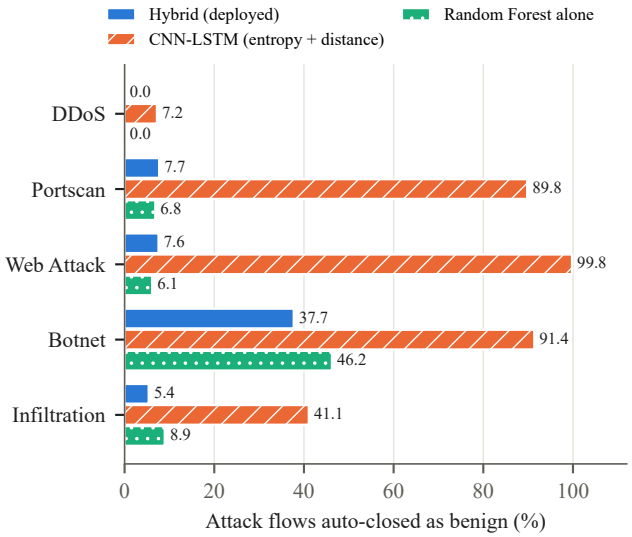


Fig. 3. Day-based split (train Monday–Wednesday, evaluate Thursday–Friday): share of each unseen attack class auto-closed as benign.

class (Fig. 3), Botnet is the clear weak spot, with 37.7% of its flows closed unseen.

F. System behaviour

While replaying traffic, the service needs about 0.55 ms of CPU time per flow, forest included. One automated test walks through a whole working session: traffic is replayed, an analyst records verdicts, similar incidents are retrieved, an administrator verifies feedback, a candidate is retrained and checked, then approved, deployed and rolled back, and every step must appear in the audit log. The same test confirms that no analyst-facing response ever contains a true label.

VII. Discussion

What the numbers do and do not show. With leakage removed, every model we trained passes 99.7% binary F1 on a random split, and the forest edges out the network. On such a split the contest is effectively over before it starts; the differences that matter for an operator only appear when the traffic contains something the model has not met before.

Confidence is not safety. The network's own scores let up to 97% of a novel attack through, and how badly it failed changed from run to run. The single change that helped most was to pair it with a model built on entirely different principles and to send their disagreements to a person. This is in line with earlier evidence that several independent models express doubt more reliably than one network does [13].

Caveats. The combined rule was chosen after we had seen the withheld-family results, so those results flatter it. The test partition was used only once, but it contains no attack type that the models had not met in training. Each withheld family was run with a single seed. Our comparison with the reference paper differs in data release and preprocessing. The two classes with fewer than 100 records cannot be scored meaningfully. Finally, the analyst workflow was exercised by the authors rather than by working security staff.

Limitations. A model that judges one flow at a time cannot see attacks that only make sense across many flows, such as SSH password guessing and part of the botnet traffic; features summarising a host's activity over a time window would be needed. Retraining currently updates the network but not the forest. The generator's samples are unrealistic, so generation that respects integer-valued features, or the VAE seeding of the reference design, is worth trying. Login and access control in the prototype are simulated.

VIII. Conclusion

This work set out to build an intrusion detector that knows when to hand a decision to a person. Under careful leakage control, our CNN-LSTM comes close to strong tree baselines and scores above the figures published for the reference design, yet its own sense of uncertainty breaks down on attacks it has never met. Adding a Random Forest as a second opinion fixes most of that: only 1.48% of traffic needs a human, that share contains 98.9% of the network's errors, and most flows from withheld attack families end up with an analyst instead of being closed silently. Analyst knowledge flows back into the model only through verified, threshold-gated and administrator-approved retraining that can be undone. Next steps are features that summarise many flows per host, retraining the forest alongside the network, and testing on CSE-CIC-IDS2018 to see how the system copes with a different network.

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